AGILE LAB FAT EXAM

Deepak C 23MIS0374

Set -01

Food Order Processing System

Problem Statement: Ensure food orders are placed and processed correctly without errors

Tasks:

1. Develop validation logic

2. Write JUnit tests for transaction success/failure

3. Implement CI/CD pipeline with automated testing

> Set up version control using GitHub

> Configure CI pipeline (e.g., Jenkins/GitHub Actions) to trigger on code commits Automate build process using Maven > Integrate automated unit tests in pipeline Configure CD pipeline for automatic deployment to staging/production

4. Deploy services on Docker and Kubernetes

> Containerize application using Docker (write Dockerfile)

> Build and push Docker image to container registry (Docker Hub)

Create Kubernetes deployment and service YAML files

OrderProcessor.java

```
import java.util.*;


public class OrderProcessor {
```

```
    public static String placeOrder(Map<String, Object> order) {
        if (order == null) {
            return "FAILED: Order is null";
        }
        if (!order.containsKey("itemName") || order.get("itemName")
== null) {
            return "FAILED: Item name is missing";
        }
        if (!order.containsKey("quantity") || (int)
order.get("quantity") <= 0) {
            return "FAILED: Invalid quantity";
        }
        if (!order.containsKey("price") || (double) order.get("price")
<= 0.0) {
            return "FAILED: Invalid price";
        }
        return "SUCCESS: Order placed for " +
order.get("itemName");
    }

    public static double calculateTotal(int quantity, double price) {
        if (quantity <= 0 || price <= 0.0) {
            return -1.0;
        }
        return quantity * price;
```

```
    }

    public static void main(String[] args) {
        Map<String, Object> order = new HashMap<>();
        order.put("itemName", "Laptop");
        order.put("quantity", 2);
        order.put("price", 50000.0);

        System.out.println(placeOrder(order));
        System.out.println("Total: " + calculateTotal(2, 50000.0));
    }
}
```

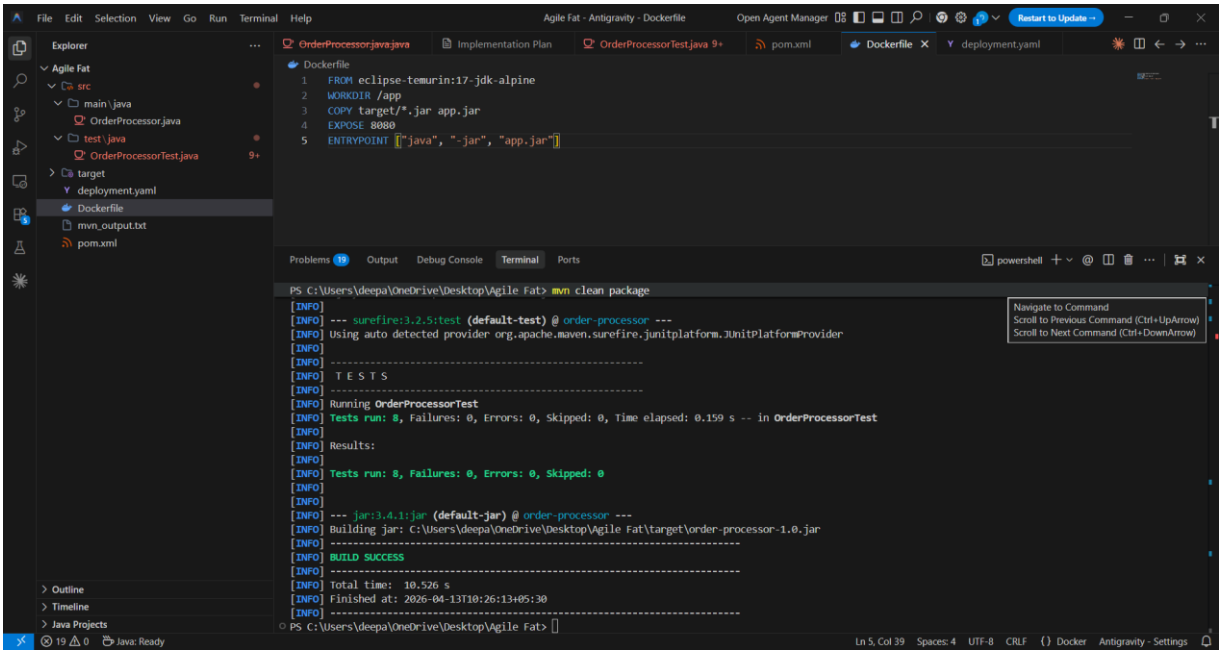
OrderProcessorTest.java

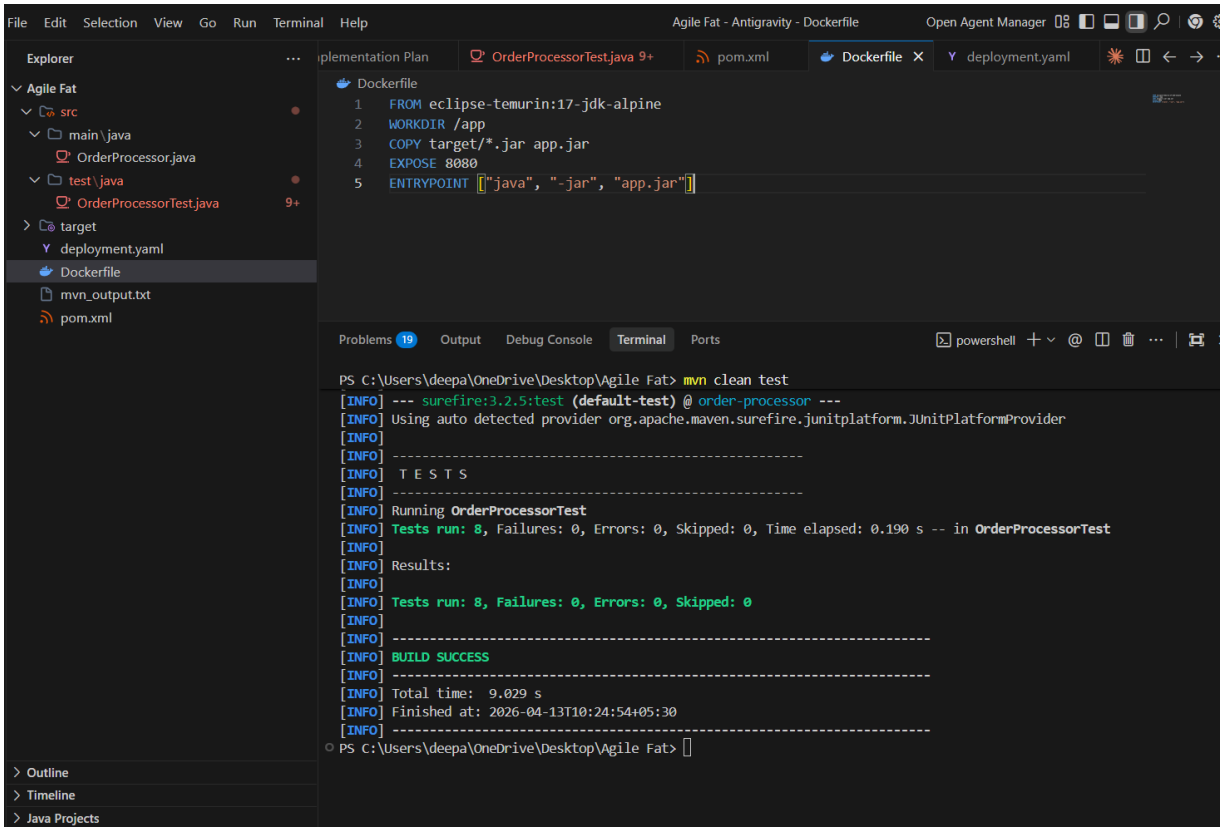
```
import static org.junit.jupiter.api.Assertions.*;
import org.junit.jupiter.api.Test;
import java.util.*;

public class OrderProcessorTest {

    // ── SUCCESS TESTS ──

    @Test
    void testOrderPlacedSuccessfully() {
        Map<String, Object> order = new HashMap<>();
```

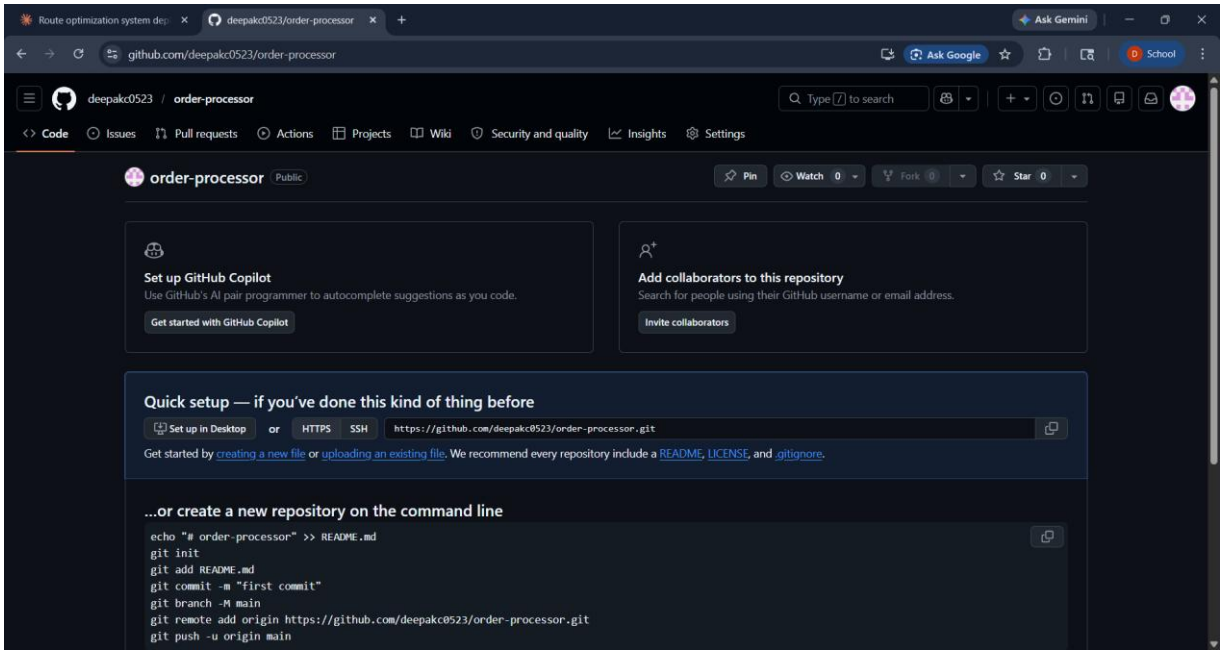
```
        order.put("itemName", "Laptop");
        order.put("quantity", 2);
        order.put("price", 50000.0);
        assertEquals("SUCCESS: Order placed for Laptop",
                OrderProcessor.placeOrder(order));
    }

    @Test
    void testCalculateTotalCorrect() {
        assertEquals(100000.0, OrderProcessor.calculateTotal(2,
50000.0));
    }

    @Test
    void testSingleItemOrder() {
        Map<String, Object> order = new HashMap<>();
        order.put("itemName", "Phone");
        order.put("quantity", 1);
        order.put("price", 20000.0);
        assertTrue(OrderProcessor.placeOrder(order).startsWith("SUC
CESS"));
    }

    // ── FAILURE TESTS ──
```

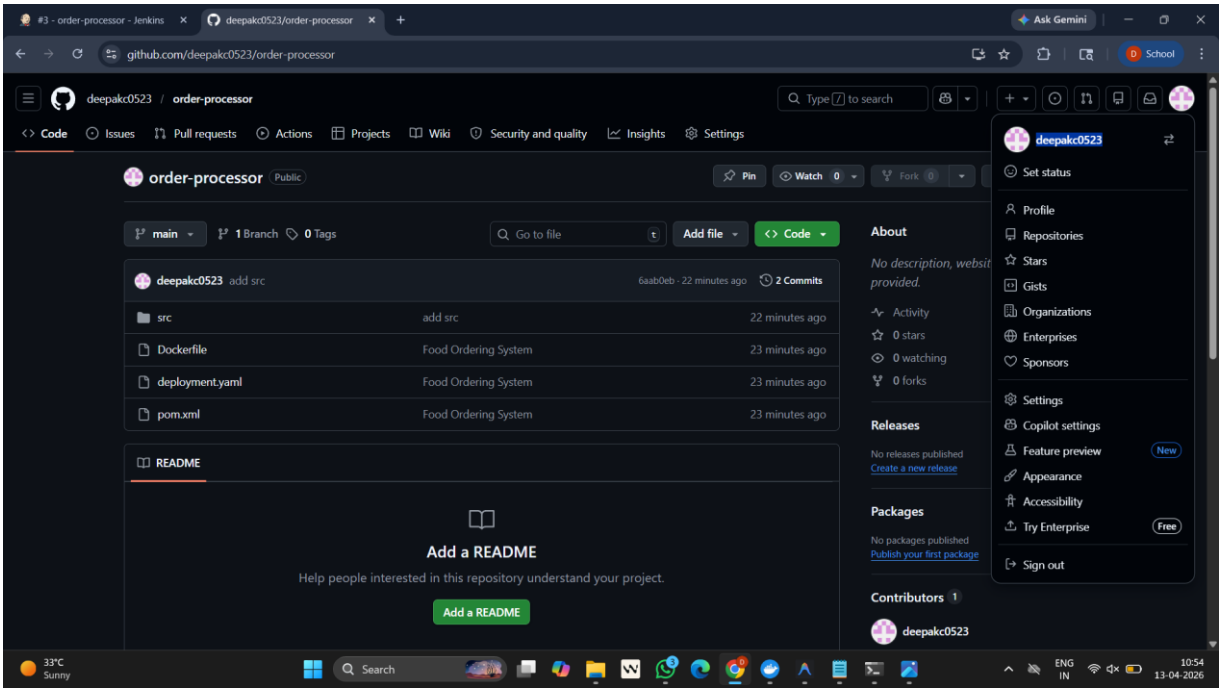
```
    @Test
    void testNullOrderFails() {
        assertEquals("FAILED: Order is null",
                OrderProcessor.placeOrder(null));
    }

    @Test
    void testMissingItemNameFails() {
        Map<String, Object> order = new HashMap<>();
        order.put("quantity", 2);
        order.put("price", 500.0);
        assertTrue(OrderProcessor.placeOrder(order).startsWith("FAI
LED"));
    }

    @Test
    void testZeroQuantityFails() {
        Map<String, Object> order = new HashMap<>();
        order.put("itemName", "Tablet");
        order.put("quantity", 0);
        order.put("price", 15000.0);
        assertTrue(OrderProcessor.placeOrder(order).startsWith("FAI
LED"));
```

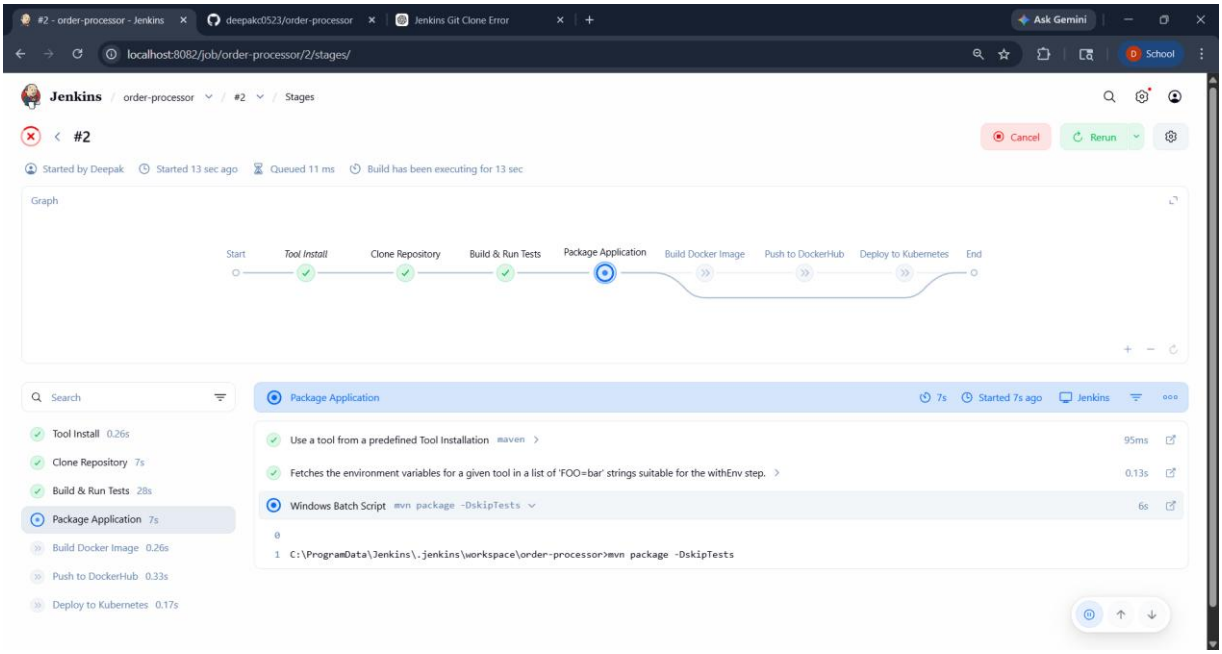
```
    }

    @Test
    void testNegativePriceFails() {
        Map<String, Object> order = new HashMap<>();
        order.put("itemName", "Watch");
        order.put("quantity", 1);
        order.put("price", -100.0);
        assertTrue(OrderProcessor.placeOrder(order).startsWith("FAI
LED"));
    }

    @Test
    void testInvalidTotalReturnsNegative() {
        assertEquals(-1.0, OrderProcessor.calculateTotal(0, 500.0));
    }
}
```

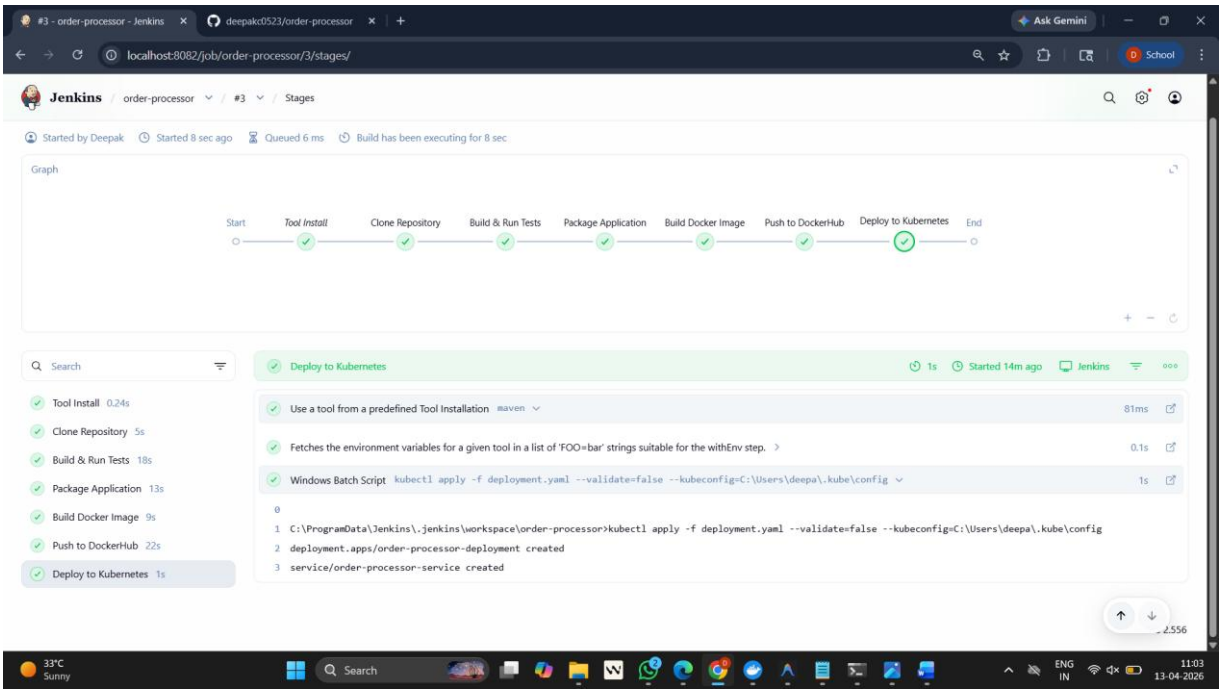

File Edit Selection View Go Run Terminal Help
Agile Fat - Antigravity - Dockerfile
Open Agent Manager
Restart to Update
Explorer
Agile Fat
src
main\java
OrderProcessor.java
test\java
OrderProcessorTest.java
target
deployment.yaml
Dockerfile
mvn_output.txt
pom.xml
Implementation Plan
OrderProcessorTest.java 9+
Dockerfile
deployment.yaml
1 FROM eclipse-temurin:17-jdk-alpine
2 WORKDIR /app
3 COPY target/*.jar app.jar
4 EXPOSE 8080
5 ENTRYPOINT ["java", "-jar", "app.jar"]
Problems 19
Output
Debug Console
Terminal
Ports
powershell
PS C:\Users\deepa\OneDrive\Desktop\Agile Fat> mvn clean package
[INFO] --- surefire:3.2.5:test (default-test) @ order-processor ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO] T E S T S
[INFO] Running OrderProcessorTest
[INFO] Tests run: 8, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.159 s -- in OrderProcessorTest
[INFO] Results:
[INFO] Tests run: 8, Failures: 0, Errors: 0, Skipped: 0
[INFO] --- jar:3.4.1:jar (default-jar) @ order-processor ---
[INFO] Building jar: C:\Users\deepa\OneDrive\Desktop\Agile Fat\target\order-processor-1.0.jar
[INFO] BUILD SUCCESS
[INFO] Total time: 10.526 s
[INFO] Finished at: 2026-04-13T10:26:13+05:30
Navigate to Command
Scroll to Previous Command (Ctrl+UpArrow)
Scroll to Next Command (Ctrl+DownArrow)
Outline
Timeline
Java Projects
Java: Ready
Ln 5, Col 39
Spaces: 4
UTF-8
CRLF
Docker
Antigravity - Settings

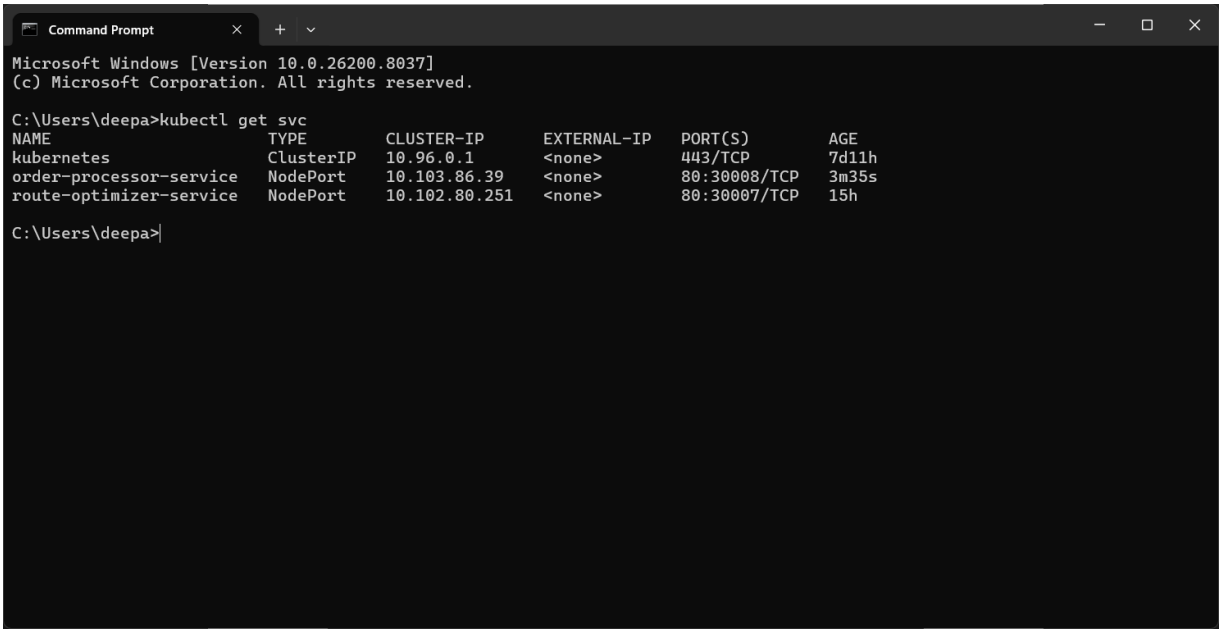

File Edit Selection View Go Run Terminal Help
Agile Fat - Antigravity - Dockerfile
Open Agent Manager
Explorer
Agile Fat
src
main\java
OrderProcessor.java
test\java
OrderProcessorTest.java
target
deployment.yaml
Dockerfile
mvn_output.txt
pom.xml
OrderProcessorTest.java 9+
pom.xml
Dockerfile
deployment.yaml
1 FROM eclipse-temurin:17-jdk-alpine
2 WORKDIR /app
3 COPY target/*.jar app.jar
4 EXPOSE 8080
5 ENTRYPOINT ["java", "-jar", "app.jar"]
Problems 19
Output
Debug Console
Terminal
Ports
powershell
PS C:\Users\deepa\OneDrive\Desktop\Agile Fat> mvn clean test
[INFO] --- surefire:3.2.5:test (default-test) @ order-processor ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO] T E S T S
[INFO] Running OrderProcessorTest
[INFO] Tests run: 8, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.190 s -- in OrderProcessorTest
[INFO] Results:
[INFO] Tests run: 8, Failures: 0, Errors: 0, Skipped: 0
[INFO] BUILD SUCCESS
[INFO] Total time: 9.029 s
[INFO] Finished at: 2026-04-13T10:24:54+05:30
Outline
Timeline
Java Projects

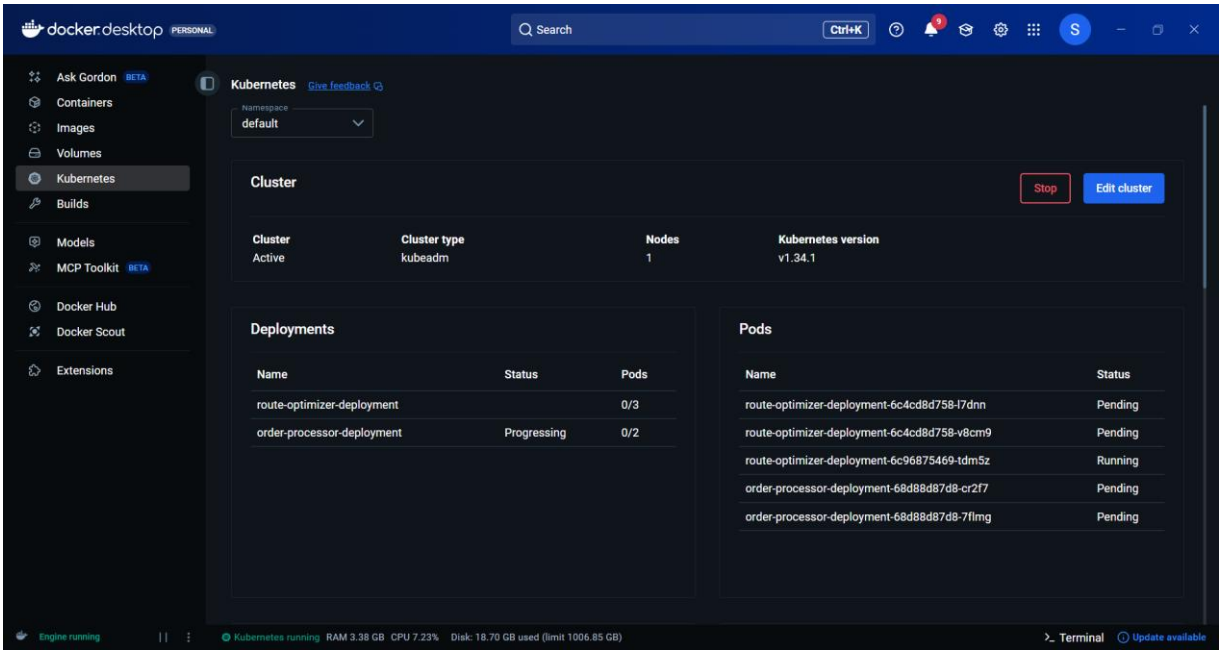

github.com/deepakc0523/order-processor
deepakc0523 / order-processor
Code
Issues
Pull requests
Actions
Projects
Wiki
Security and quality
Insights
Settings
order-processor
Public
Pin
Watch 0
Fork 0
Star 0
Set up GitHub Copilot
Use GitHub's AI pair programmer to autocomplete suggestions as you code.
Get started with GitHub Copilot
Add collaborators to this repository
Search for people using their GitHub username or email address.
Invite collaborators
Quick setup — if you've done this kind of thing before
Set up in Desktop
or
HTTPS
SSH
https://github.com/deepakc0523/order-processor.git
Get started by creating a new file or uploading an existing file. We recommend every repository include a README, LICENSE, and .gitignore.
...or create a new repository on the command line
echo "# order-processor" >> README.md
git init
git add README.md
git commit -m "first commit"
git branch -M main
git remote add origin https://github.com/deepakc0523/order-processor.git
git push -u origin main


#3 - order-processor - Jenkins
github.com/deepakc0523/order-processor
deepakc0523 / order-processor
Code
Issues
Pull requests
Actions
Projects
Wiki
Security and quality
Insights
Settings
order-processor
Public
main
1 Branch
0 Tags
Go to file
Add file
Code
deepakc0523 add src
6aab0eb · 22 minutes ago
2 Commits
src
add src
22 minutes ago
Dockerfile
Food Ordering System
23 minutes ago
deployment.yaml
Food Ordering System
23 minutes ago
pom.xml
Food Ordering System
23 minutes ago
README
Add a README
Help people interested in this repository understand your project.
Add a README
About
Activity
0 stars
0 watching
0 forks
Releases
No releases published
Create a new release
Packages
No packages published
Publish your first package
Contributors 1
deepakc0523
deepakc0523
Set status
Profile
Repositories
Stars
Gists
Organizations
Enterprises
Sponsors
Settings
Copilot settings
Feature preview
New
Appearance
Accessibility
Try Enterprise
Free
Sign out
33°C
Sunny
Search
ENG
IN
10:54
13-04-2026


#2 - order-processor - Jenkins
deepakc0523/order-processor
Jenkins Git Clone Error
Ask Gemini
localhost:8082/job/order-processor/2/stages/
School
Jenkins / order-processor / #2 / Stages
#2
Cancel
Rerun
Started by Deepak
Started 13 sec ago
Queued 11 ms
Build has been executing for 13 sec
Graph
Start
Tool Install
Clone Repository
Build & Run Tests
Package Application
Build Docker Image
Push to DockerHub
Deploy to Kubernetes
End
Search
Package Application
7s
Started 7s ago
Jenkins
Tool Install 0.26s
Clone Repository 7s
Build & Run Tests 28s
Package Application 7s
Build Docker Image 0.26s
Push to DockerHub 0.33s
Deploy to Kubernetes 0.17s
Use a tool from a predefined Tool Installation maven
95ms
Fetches the environment variables for a given tool in a list of 'FOO=bar' strings suitable for the withEnv step.
0.13s
Windows Batch Script mvn package -DskipTests
6s
0
1 C:\ProgramData\Jenkins\.jenkins\workspace\order-processor>mvn package -DskipTests


#3 - order-processor - Jenkins
deepakc0523/order-processor
Ask Gemini
localhost:8082/job/order-processor/3/stages/
School
Jenkins / order-processor / #3 / Stages
Started by Deepak
Started 8 sec ago
Queued 6 ms
Build has been executing for 8 sec
Graph
Start
Tool Install
Clone Repository
Build & Run Tests
Package Application
Build Docker Image
Push to DockerHub
Deploy to Kubernetes
End
Search
Deploy to Kubernetes
1s
Started 14m ago
Jenkins
Tool Install 0.24s
Clone Repository 5s
Build & Run Tests 18s
Package Application 13s
Build Docker Image 9s
Push to DockerHub 22s
Deploy to Kubernetes 1s
Use a tool from a predefined Tool Installation maven
81ms
Fetches the environment variables for a given tool in a list of 'FOO=bar' strings suitable for the withEnv step.
0.1s
Windows Batch Script kubectl apply -f deployment.yaml --validate=false --kubeconfig=C:\Users\deepa\.kube\config
1s
0
1 C:\ProgramData\Jenkins\.jenkins\workspace\order-processor>kubectl apply -f deployment.yaml --validate=false --kubeconfig=C:\Users\deepa\.kube\config
2 deployment.apps/order-processor-deployment created
3 service/order-processor-service created
33°C
Sunny
Search
ENG
IN
11:03
13-04-2026





The pipeline Script:

```
pipeline {
    agent any
    tools {
        maven 'maven'
```

```
}
environment {
    DOCKER_IMAGE = "sauldee/order-processor:latest"
}
stages {
    stage('Clone Repository') {
        steps {
            git branch: 'main',
            url: 'https://github.com/deepakc0523/order-processor.git'
        }
    }
    stage('Build & Run Tests') {
        steps {
            bat 'mvn clean test'
        }
        post {
            always {
                junit 'target/surefire-reports/*.xml'
            }
        }
    }
    stage('Package Application') {
        steps {
            bat 'mvn package -DskipTests'
```

```
        }
    }
    stage('Build Docker Image') {
        steps {
            bat "docker build -t %DOCKER_IMAGE% ."
        }
    }
    stage('Push to DockerHub') {
        steps {
            withCredentials([usernamePassword(
                credentialsId: 'dockerhub-creds',
                usernameVariable: 'DOCKER_USER',
                passwordVariable: 'DOCKER_PASS'
            )]) {
                bat """
                docker login -u %DOCKER_USER% -p
%DOCKER_PASS%
                docker push %DOCKER_IMAGE%
                """
            }
        }
    }
    stage('Deploy to Kubernetes') {
        steps {
```

```
                bat 'kubectl apply -f deployment.yaml --validate=false --
kubeconfig=C:\\Users\\deepa\\.kube\\config'
            }
        }
    }
}
```